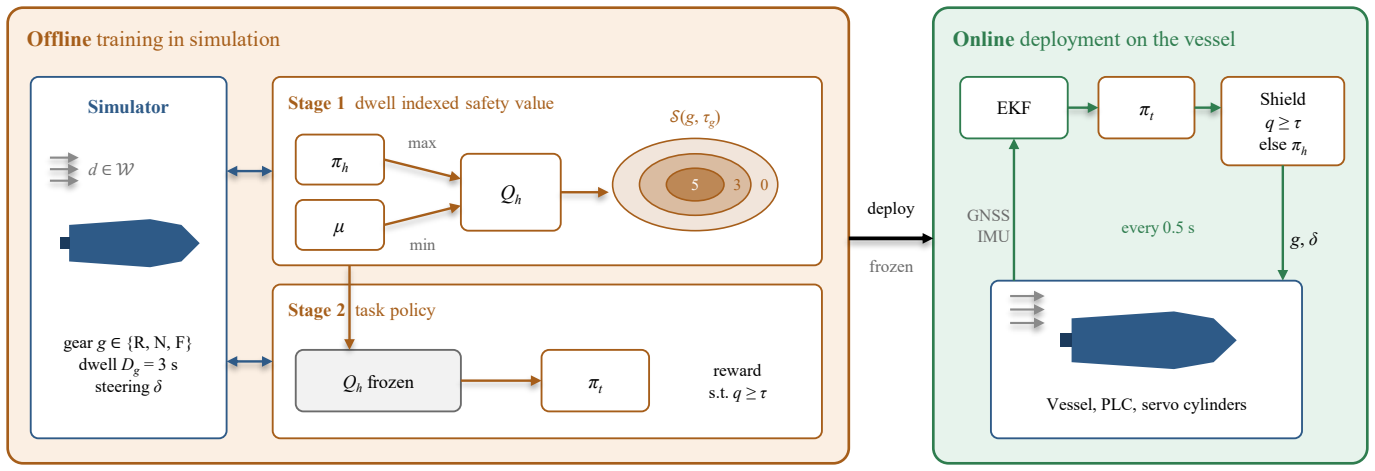

Offline training in simulation
Simulator
$d \in \mathcal{W}$
gear $g \in$ {R, N, F}
dwell $D_g = 3$ s
steering δ
Stage 1 dwell indexed safety value
π_h
μ
max
min
Q_h
$S(g, \tau_g)$
5
3
0
Stage 2 task policy
Q_h frozen
π_t
reward
s.t. $q \geq \tau$
deploy
frozen
Online deployment on the vessel
EKF
π_t
Shield
$q \geq \tau$
else π_h
GNSS
IMU
every 0.5 s
g, δ
Vessel, PLC, servo cylinders